

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 1: Radio Wave Propagation, Antennas & Frequencies

Syllabus Source: CAE Oxford (pp. 1-44) | Pass Target: >= 90% AviationExam

1. THE ELECTROMAGNETIC SPECTRUM & FREQUENCY BANDS

Radio waves travel at the speed of light ( $c = 300,000 \text{ km/s} = 3 \times 10^8 \text{ m/s}$ ). Wavelength is inversely proportional to frequency:  $\lambda \text{ (meters)} = c / \text{frequency (Hz)}$ :

| Frequency Band        | Frequency Range & Wavelength     | Aviation Navigational Aids & Communications                                               |
|-----------------------|----------------------------------|-------------------------------------------------------------------------------------------|
| Frequency Band        | Frequency Range & Wavelength     | Aviation Navigational Aids & Communications                                               |
| VLF (Very Low)        | 3 to 30 kHz (100 to 10 km)       | Submarine communications, Omega navigation (obsolete).                                    |
| LF (Low Frequency)    | 30 to 300 kHz (10 to 1 km)       | NDB (Non-Directional Beacons), commercial AM radio.                                       |
| MF (Medium Frequency) | 300 to 3,000 kHz (1 km to 100 m) | NDB, Locator beacons, maritime radio.                                                     |
| HF (High Frequency)   | 3 to 30 MHz (100 to 10 m)        | Long-range transoceanic voice communications (Sky wave).                                  |
| VHF (Very High)       | 30 to 300 MHz (10 to 1 m)        | VHF Comms (118-137 MHz), VOR (108-118 MHz), ILS Localizer (108-112 MHz).                  |
| UHF (Ultra High)      | 300 to 3,000 MHz (1 m to 10 cm)  | ILS Glide Path (329-335 MHz), DME (960-1215 MHz), SSR (1030/1090 MHz), GPS L1 (1575 MHz). |
| SHF (Super High)      | 3 to 30 GHz (10 to 1 cm)         | Airborne Weather Radar (AWR 9.3 GHz), MLS (5 GHz), Radar Altimeter (4.3 GHz).             |

2. WAVE PROPAGATION MODES & MAXIMUM RANGE

- Ground Wave (Surface Wave): Follows the curvature of the Earth. Dominant in LF and MF bands (NDB). Range is greatest over seawater (high conductivity) and lowest over dry sand/ice.
- Sky Wave (Ionospheric Reflection): Reflected by ionosphere layers (D, E, F1, F2). Dominant in HF band. Subject to skip distance, skip zones, and severe night effect fading.
- Space Wave (Line-of-Sight): Straight-line propagation in VHF, UHF, and SHF bands. Limited by atmospheric refraction and Earth curvature: Maximum Theoretical Range (NM) =  $1.23 \times (\sqrt{h_{\text{transmitter}}} + \sqrt{h_{\text{receiver}}})$  [heights in feet]!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): VHF VOR Line-of-Sight Range Calculation**

Problem: A VOR station is installed at 400 ft MSL. An airliner cruises at FL 250 (25,000 ft). Find the theoretical maximum radio line-of-sight reception range in nautical miles. Step 1: Formula:  $\text{Range (NM)} = 1.23 \times \sqrt{h_{tx}} + 1.23 \times \sqrt{h_{rx}}$ . Step 2: Transmitter height term:  $\sqrt{400} = 20 \rightarrow 1.23 \times 20 = 24.6$  NM. Step 3: Aircraft height term:  $\sqrt{25,000} = 158.11 \rightarrow 1.23 \times 158.11 = 194.5$  NM. Step 4: Total Range =  $24.6 + 194.5 = 219.1$  NM! Quick Rule: If the ground beacon is at sea level (0 ft), range depends only on aircraft altitude:  $1.23 \times \sqrt{\text{Altitude}}$ .

**3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)****[\*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: VHF Maximum Line-of-Sight Radio Range**

SCENARIO (Fundamental AviationExam Question): An aircraft is cruising at FL 360 (36,000 ft). A VOR ground station antenna is located at an elevation of 900 ft MSL. QUESTION: What is the maximum theoretical range in Nautical Miles at which the aircraft can receive the VOR signal? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Identify Transmitter and Receiver Heights in feet: - Aircraft Height ( $h_{rx}$ ) = 36,000 ft - VOR Ground Antenna Height ( $h_{tx}$ ) = 900 ft Step 2: Recall the Master VHF Line-of-Sight Range Formula: - Formula:  $\text{Range (NM)} = 1.23 \times [\sqrt{h_{tx}} + \sqrt{h_{rx}}]$  Step 3: Calculate the square roots of each height: -  $\sqrt{900} = 30$  -  $\sqrt{36,000} = 189.74$  - Sum of square roots =  $30 + 189.74 = 219.74$  Step 4: Multiply by the atmospheric refraction constant (1.23): -  $\text{Range (NM)} = 1.23 \times 219.74 = 270.28$  NM (~270 Nautical Miles)! FINAL ANSWER: The maximum theoretical VOR reception range is approximately 270 Nautical Miles.

**[!] EXAM TRAP (AVIATIONEXAM) - The 1.05 vs 1.23 Range Constant Trap**

- If both antennas are optical line-of-sight in a vacuum: constant = 1.05. - In the Earth's atmosphere, radio waves BEND SLIGHTLY downwards due to atmospheric refraction: constant = 1.23 (or 1.25)! - EASA exams universally use 1.23:  $\text{Range} = 1.23 \times (\sqrt{h_1} + \sqrt{h_2})$ .

**4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)****Q1: In which frequency band does the Instrument Landing System (ILS) LOCALIZER operate?**

- [A]: UHF
- [B]: VHF (108.10 MHz to 111.95 MHz, odd tenths)
- [C]: SHF
- [D]: LF

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. The ILS Localizer operates in the VHF band between 108.10 and 111.95 MHz on odd-tenth frequencies (e.g. 109.10, 109.30). The Glide Path operates in UHF.

**Q2: Over which surface type does an LF/MF Ground Wave travel with the GREATEST range and minimum signal loss?**

- [A]: Dry desert sand
- [B]: SEAWATER (due to high electrical conductivity)
- [C]: Dense forest
- [D]: Mountainous rocky terrain

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. Seawater has the highest electrical conductivity of any natural Earth surface, producing minimum ground attenuation and maximum surface wave range.

**Q3: What is the wavelength of a radar operating on a frequency of 10 GHz (10,000 MHz)?**

- [A]: 3 meters
- [B]: 3 CENTIMETERS ( $\lambda = 300,000,000 \text{ m/s} / 10,000,000,000 \text{ Hz} = 0.03 \text{ m} = 3 \text{ cm}$ )
- [C]: 30 cm
- [D]: 30 meters

**[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation**

CORRECT: [B]. Formula:  $\lambda = c / f$ . For 10 GHz:  $3 \times 10^8 / 10 \times 10^9 = 0.03 \text{ m} = 3 \text{ cm}$  (standard Airborne Weather Radar X-band).